

VELIKIYE LUKI, Russian Federation

WMO: 264770

Lat: 56.350N

Lon: 30.617E

Elev: 103

StdP: 100.09

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-19.2	-25.4	0.4	-22.2	-22.0	0.5	-18.5	8.8	-5.3	7.8	-1.7	0.7	100	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	28.2	19.5	26.6	18.9	25.1	18.1	21.0	25.9	20.0	24.8	19.0	23.3	2.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.2	23.3	18.3	13.3	21.9	17.3	12.5	21.1	61.2	25.7	57.6	24.9	54.2	23.3	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.5	5.7	-27.2	30.6	4.3	1.5	-30.3	31.7	-32.8	32.5	-35.2	33.3	-38.3	34.4		
(5)			-27.4	22.5	4.2	1.0	-30.4	23.2	-32.9	23.8	-35.3	24.3	-38.3	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-5.8	-4.5	-0.6	7.0	12.3	16.5	18.1	16.7	11.5	5.9	0.4	-3.3		
	DBStd	9.70	6.34	6.59	4.65	4.80	4.64	3.79	3.10	3.02	3.84	4.14	5.24	6.00		
	HDD10.0	2237	489	407	329	117	29	2	0	0	28	136	288	411		
	HDD18.3	4499	747	641	588	341	193	80	42	70	205	385	538	669		
	CDD10.0	864	0	0	0	27	101	196	252	207	74	9	0	0		
	CDD18.3	86	0	0	0	0	6	24	36	18	1	0	0	0		
	CDH23.3	767	0	0	0	4	69	203	291	182	18	0	0	0		
	CDH26.7	136	0	0	0	0	10	30	63	32	1	0	0	0		
(14) Wind	WSAvg	2.3	2.7	2.8	2.8	2.4	2.2	2.0	1.8	1.7	1.9	2.4	2.5	2.7		
(15) Precipitation	PrecAvg	596	33	30	31	34	54	74	82	70	57	50	42	42		
	PrecMax	830	84	71	70	96	126	183	213	204	168	110	108	68		
	PrecMin	437	5	3	4	2	9	31	18	5	11	5	0	16		
	PrecStd	100	16	15	15	23	28	34	38	42	35	26	21	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	6.0	13.4	24.0	28.1	29.4	30.8	29.5	25.7	18.8	11.5	8.5		
		MCWB	4.0	3.5	6.8	13.1	18.4	20.0	21.4	19.4	17.9	13.5	10.4	7.1		
	2%	DB	3.2	4.3	9.6	21.2	25.3	27.4	28.5	27.2	22.4	15.2	9.7	5.9		
		MCWB	2.5	2.3	4.9	12.4	16.8	18.9	20.5	19.0	16.6	11.8	8.5	5.1		
	5%	DB	2.3	3.2	7.2	18.2	22.9	25.8	26.5	25.3	19.9	13.2	8.5	4.6		
		MCWB	1.4	1.8	3.7	11.4	15.1	18.3	19.5	18.2	15.1	10.9	7.5	4.0		
	10%	DB	1.5	2.4	5.3	15.4	20.6	24.0	24.6	23.4	17.7	11.9	7.0	3.1		
		MCWB	0.7	1.3	3.0	9.7	14.2	17.6	18.7	17.2	13.9	10.2	6.1	2.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.0	8.2	14.7	19.7	21.5	23.8	21.4	18.7	14.4	10.7	7.3		
		MCDB	4.7	5.4	12.2	21.7	26.3	26.7	29.5	26.6	23.6	17.8	11.2	8.0		
	2%	WB	2.7	2.7	5.7	13.1	17.8	20.4	21.7	19.9	17.1	12.4	8.7	5.2		
		MCDB	3.1	3.6	7.9	18.9	23.1	25.4	26.1	25.1	21.5	14.1	9.5	5.8		
	5%	WB	1.6	2.1	4.4	11.8	16.5	19.3	20.5	18.9	15.8	11.4	7.5	4.0		
		MCDB	2.1	3.0	6.4	17.2	21.2	23.7	24.9	23.7	19.1	13.1	8.4	4.6		
	10%	WB	0.8	1.5	3.2	10.5	15.1	18.3	19.2	18.0	14.7	10.2	6.1	2.5		
		MCDB	1.4	2.2	5.1	15.1	19.3	22.6	23.3	22.1	17.1	11.8	6.9	3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	6.5	7.5	9.7	10.7	10.0	9.6	10.5	9.0	6.4	4.3	4.0		
		MCDBR	3.6	5.0	9.9	14.4	14.2	12.6	13.0	13.9	13.0	9.0	5.0	3.6		
		MCWBR	3.5	3.9	6.1	7.0	6.4	5.6	5.9	6.4	7.1	6.0	4.5	3.3		
		MCWBR	3.5	3.9	7.6	13.3	12.3	11.0	11.4	11.8	11.8	8.2	4.7	3.4		
(40) Clear-Sky Solar Irradiance	taud	taub	0.288	0.322	0.347	0.397	0.386	0.376	0.401	0.408	0.369	0.348	0.328	0.288		
		taud	2.146	2.117	2.169	2.228	2.340	2.389	2.337	2.309	2.394	2.383	2.245	2.162		
		Ebn at Noon	586	712	800	816	853	867	833	798	781	695	538	485		
		Edh at Noon	58	89	109	120	114	110	114	110	88	70	54	45		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.50	1.34	2.78	3.92	5.13	5.46	5.28	4.21	2.69	1.23	0.42	0.29		
		RadStd	0.11	0.19	0.34	0.47	0.49	0.48	0.51	0.43	0.26	0.26	0.07	0.08		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page